Density Plots

- A Density Plot (Kernel Density Plot) shows how data is distributed across values
- It is a smooth curve, unlike a histogram which uses bars
- Higher peaks = more data points in that area
- Helps to understand the shape of data distribution easily
- Useful for comparing multiple datasets

In Pandas:

- Use pandas.DataFrame.plot.density()
- Or use pandas.DataFrame.plot.kde() (both do the same thing)

Sample code using Pandas

```
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

import pandas as pd

iris = load_iris(as_frame=True)

df = iris.frame

# Creating the visulization

sepal_series = df['sepal length (cm)']

sepal_series.plot.density(bw_method='scott', color='blue', linestyle='-', linewidth=2)

plt.show()
```

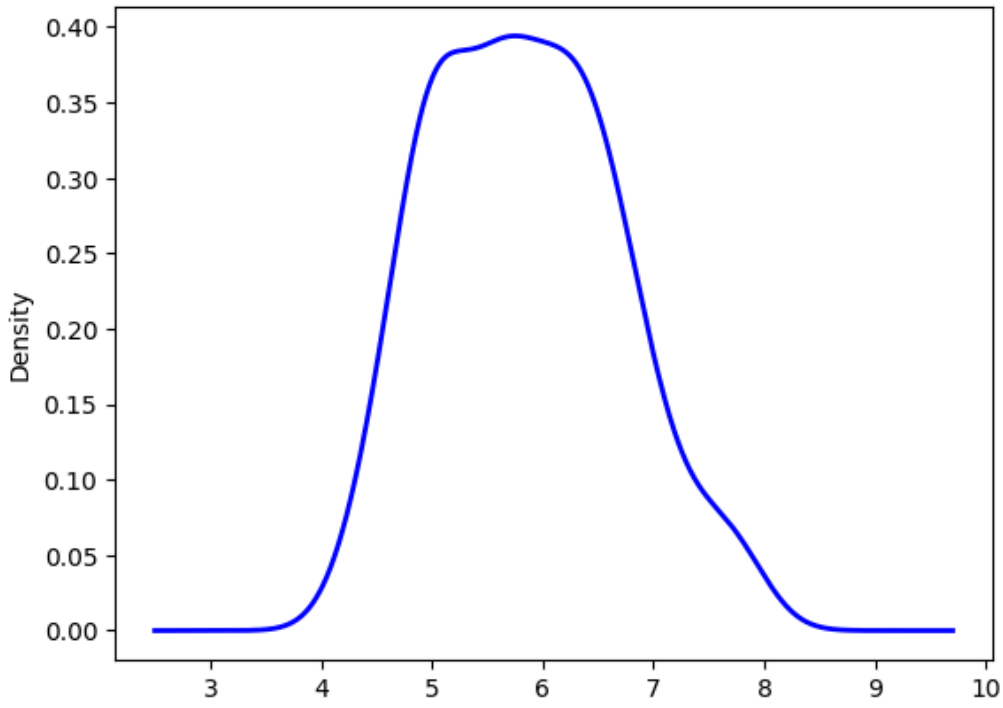

Fig 1: Density Plot for Sepal_length

KDE Method in Pandas

sepal_series.plot.kde(bw_method='scott', color='blue', linestyle='-', linewidth=2)

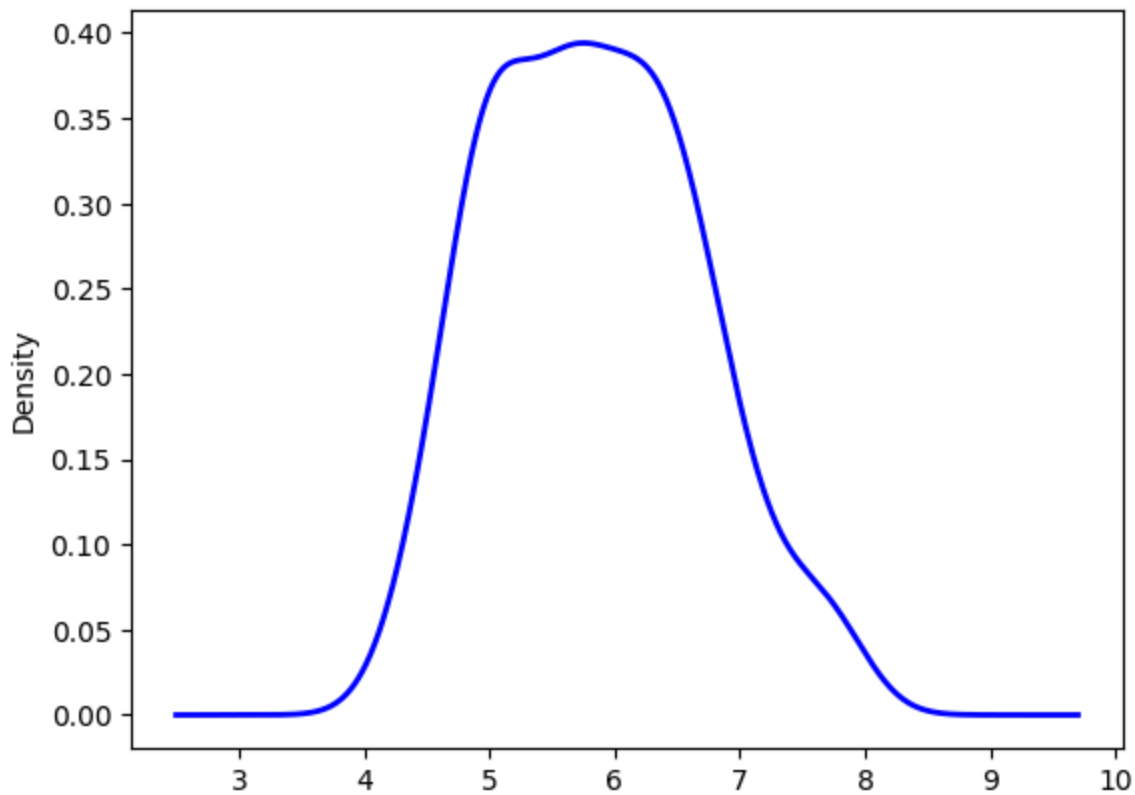

Fig 2: The KDE Method for Sepal length

Why we use a Density Plot (simple):

- It helps you see where most values are present
- It shows data as a smooth line, so it looks clean and easy
- You can quickly understand data shape (like where it peaks)

For Example in the above plotted image Fig 1

- The peak = most common value
- The sides = rare values
- Your data is mostly centered in the middle

Counter plot
A Contour Plot shows data in 3D form on a 2D surface

- It uses lines (or colors) to connect points with the same value
- Think of it like a map showing heights (same height = same line)

Advantages

- Shows 3D information in simple 2D form
- Helps understand patterns and relationships
- Easy to spot high and low regions
- Good for continuous data

Disadvantages

- Does not show exact values clearly
- Needs proper data (grid/continuous data)

Sample code

```
import numpy as np
```

```
import pandas as pd

import matplotlib.pyplot as plt

from scipy.stats import gaussian_kde

from sklearn.datasets import load_iris



iris = load_iris()
```

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)



x = df['sepal length (cm)']

y = df['petal length (cm)']



xy = np.vstack([x, y])

kde = gaussian_kde(xy)



x_grid, y_grid = np.mgrid[x.min():x.max():100j, y.min():y.max():100j]

z_grid = kde(np.vstack([x_grid.flatten(),
y_grid.flatten()])).reshape(x_grid.shape)



plt.figure(figsize=(10, 7))



contour_filled = plt.contourf(x_grid, y_grid, z_grid, levels=20,
cmap='magma')

plt.contour(x_grid, y_grid, z_grid, levels=20, colors='white', alpha=0.1)



plt.colorbar(contour_filled)

plt.scatter(x, y, s=3, color='white', alpha=0.5)



plt.title('2D Density Contour')

plt.xlabel('Sepal Length (cm)')

plt.ylabel('Petal Length (cm)')
```

```
plt.show()
```

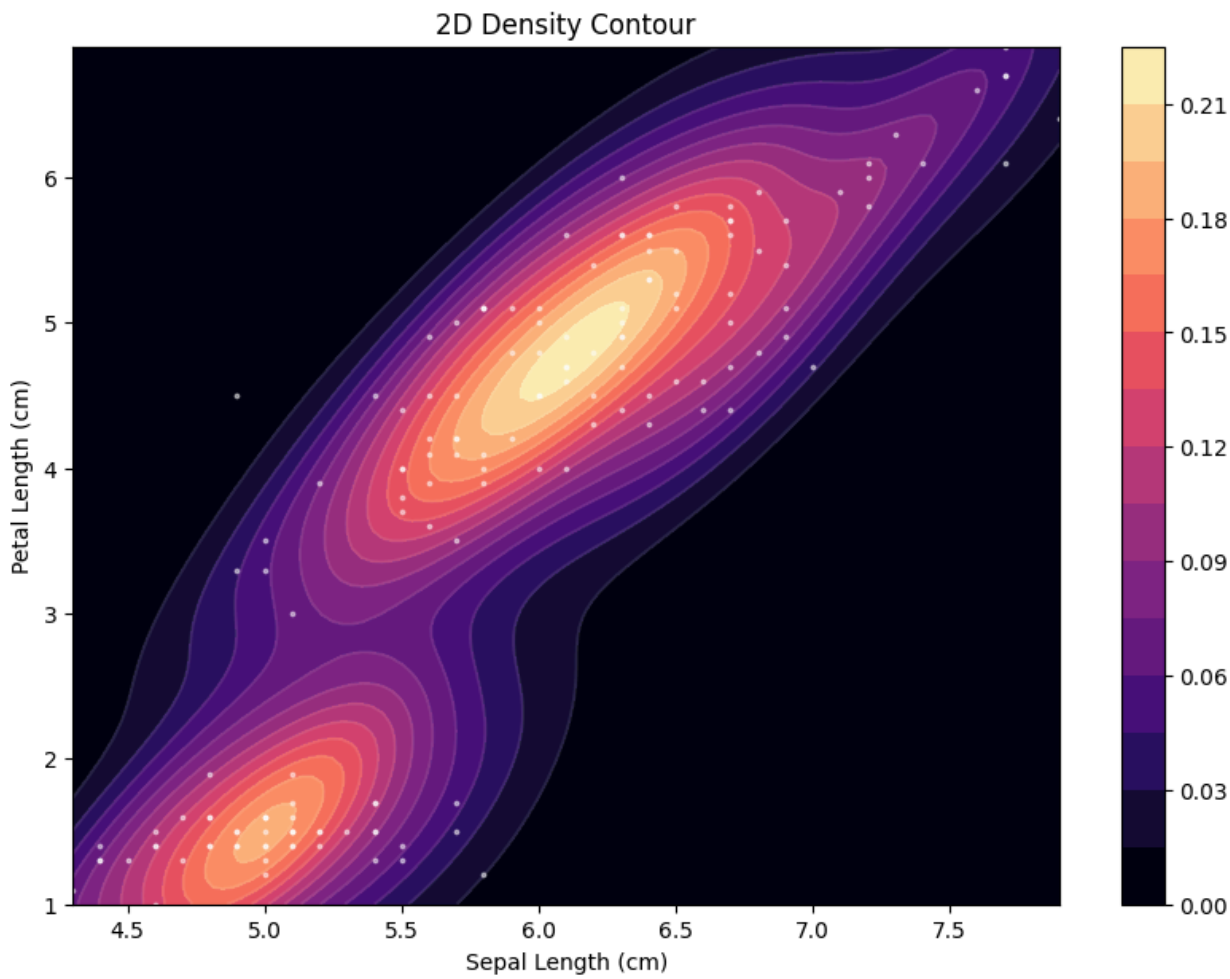

Fig 3 : counter plot for speal length and petal length